Interpretable modeling of articulatory temporal dynamics from real-time MRI for phoneme recognition

Jay Park, Hong Nguyen, Sean Foley, Jihwan Lee, Yoonjeong Lee, Dani Byrd, Shrikanth Narayanan
USC Ming Hsieh Department of Electrical and Computer Engineering

Introduction

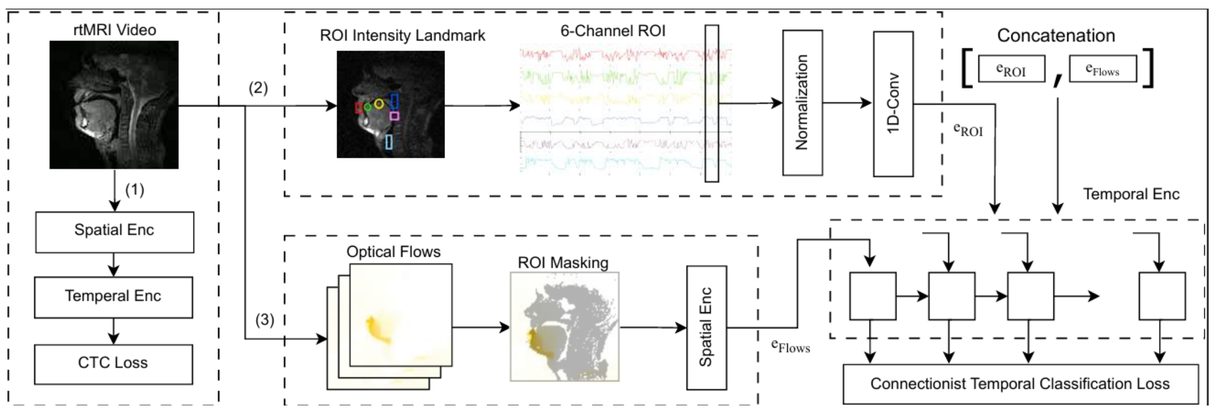
- **Real-time MRI:** A modality used to capture frames of the vocal tract during speech.
- **Optical Flow:** Representation of horizontal and vertical motion between consecutive frames.
- **Vocal Tract Regions of Interest:** Time series of average pixel intensity in manually selected articulator regions.
- **Electromagnetic Articulography (EMA):** Sensors detecting vocal tract dynamics through electromagnetic induction of coils moving around a magnetic field.

Motivation

- Real-time MRI has high dimensionality and limited interpretability.
- EMA has better interpretability but struggles to capture velum, tongue root and larynx.
- Preprocessing real-time MRI provides informative articulatory features while capturing all parts of the vocal tract.

Feature Comparison

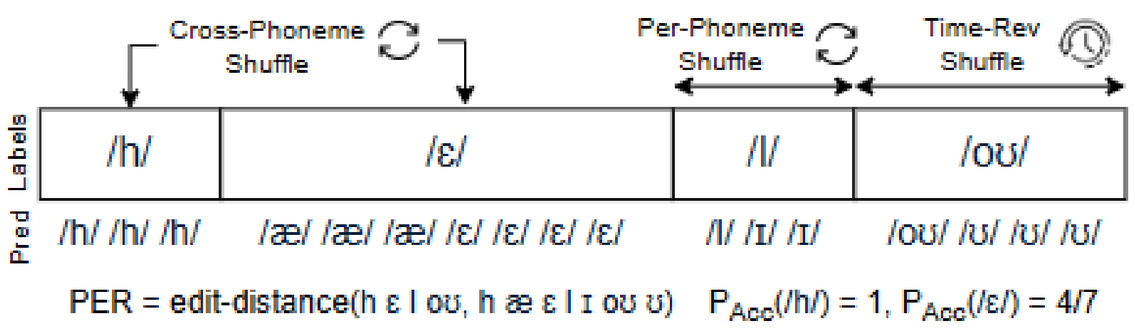
- Separate models were created for raw video, optical flow, and vocal tract regions of interest.
- Models capture spatial and temporal features before phoneme classification at each time step.



Temporal Fidelity Tests

The following temporal fidelity tests are performed to reveal the role of temporal dynamics:

1. **Cross-phoneme shuffle**: Phoneme order is randomly shuffled while maintaining frame order within phonemes.
2. **Per-phoneme shuffle**: Frame order within each phoneme is randomly shuffled.
3. **Time-reversal**: Frame order within each phoneme is reversed.
4. **Upsampling**: Frames are upsampled by a factor of two.
5. **Downsampling**: Frames are downsampled by a factor of two.



Results

rtMRI Features	Spatial Enc	Temp Enc	PER↓	Phone Acc↑ Top-1	Top-3
Raw Video	ViT-B ❄	LSTM + Conformer	0.49	N/A	
Uni-Feature					
Raw Video	ViT-B	Mamba	DIV	-	
	ResNet	Mamba	0.37	11.7	67.6
	ResNet	LSTM	0.38	12.8	**67.7**
Flows	ResNet	Mamba	0.41	11.2	67.5
	ResNet	LSTM	0.43	8.6	44.4
6C-ROI	Conv1D	LSTM	0.53	12.2	49.9
	Conv1D	Mamba	0.61	9.2	48.9
Multi-Feature					
6C-ROI + Flows			0.35	15.1	66.5
6C-ROI + Raw Video			**0.34**	14.8	64.1
Raw Video + Flows			0.39	**15.3**	66.4

Phoneme Error Rate of different sets of features

rtMRI Features	Cross-Phon.	Per-Phon.	Time-Rev.	Up Samp.	Down Samp.
Raw Vid.	0.28	**0.21**	**0.25**	**0.30**	0.20
Flows	**0.48**	0.04	0.07	0.22	**0.24**
6C-ROI	0.07	0.15	0.12	0.02	0.03

Phoneme Error Rate increase following temporal fidelity tests

LA	TT	TD	VL	TR	LX
0.13	**0.15**	0.10	0.03	0.06	0.08

Phoneme Error Rate increase upon removing articulators from regions of interest

Conclusion

- Solid multi-feature model performance demonstrates the complementary nature of video, optical flow, and vocal tract ROI.
- Temporal fidelity tests confirm reliance on within-phoneme dynamics.
- Articulators are complementary, with strong contributions from bilabial and tongue tip action.

References

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Contact: jaypark@usc.edu